GEBZE TECHNICAL UNIVERSITY
Computer Engineering Department
CSE 396 Project Proposal

Çılgın Kurye

Multiplayer Courier Game and Real-Time LED City Board

Instructor: Erkan Zergeroğlu **Company:** Board&Byte **Group:** Group-8 **Date:** 16 October 2026

Name	Student No.	Name	Student No.
Tuğberk Akbulut	230104004070	Berk Andaçgil	230104004066
Abdullatif Yılmaz	230104004031	Hasan Hüseyin Özdemir	230104004048
Hasan Alperen Elmas	1801042604	Muhib Taha Boy	230104004165
Batuhan Yanar	220104004059	Selim Barut	220104004046
Ayşe Poyraz	230104004085	Umutcan Ocak	220104004903

1. Summary

Çılgın Kurye is a multiplayer motorcycle courier game. The game is played on the players' own phones with a web browser. Up to 8 players can play the game at the same time. The whole game server runs on a single computer. Each player controls a motorcycle in a 3D city and delivers packages from the team's store to delivery points. Also, there is no map on the phone screen. All players have to watch a physical city board (60×60 cm) on the table to know where to deliver packages. All motorcycles, delivery points and obstacles are shown in real time on this board with addressable LED strips.

The main part of the project is the admin computer. From this computer, the admin can change the game parameters in real time and observe the results. The admin sees the same change in 3 places: the 3D simulation, the LED board, and the live graphs on the admin screen. The computer, the phones and the board communicate via Wi-Fi.

2. Game Description

2.1 Teams and goal

There are 4 teams at the 4 corners of the city, and there is a store building at each team corner. Each team has at most 2 players, so a game has at most 8 players. If there are not enough players, AI bots will take the empty places. Each team has a color (blue, green, purple or yellow), and the 2 players of a team get a light and a dark tone of this color. The tone of the player is used for the motorcycle in 3D, the LED dot on the board and the delivery point. All games take at most 5 minutes, and the team with more delivery points wins the game.

2.2 Delivery cycle

The player starts from the team's store and takes a package when entering the store area. Then the server computes and chooses a delivery point, and this point blinks on the board in the tone of the player. The player goes to the delivery point and delivers the package on the phone. After the delivery, the player goes back to the store to take the next package.

2.3 Driving and controls

Driving is free and the player can change lanes and pass obstacles such as cars and traffic cones. There is no reverse, so the player makes a U turn by steering. Player uses buttons on the phone screen for throttle, brake, left and right.

There is no map on the phone. Only a simple compass arrow shows the direction to the delivery point, or to the store on the way back. It shows no distance and no roads. So every player look at the physical map to find where to deliver the package.

When a player presses the "Find me" button, the LED of this player flashes white on the board for 1 second. All players can see each other in the 3D city and interact with each other via collisions.

2.4 Collisions and falls

When 2 players collide at high speed, both of them fall and wait for a short penalty time. There are also NPC cars on the roads that makes a real traffic. The player must pass between the cars. Hitting a car, building or an obstacle makes the motorcycle fall. The penalty times and traffic/obstacle density are admin parameters.

2.5 Delivery point selection

The delivery point is not fixed. Each time a player takes a package, the server chooses a delivery point with a selection algorithm. The algorithm considers how long it will take to reach the point from the team store with the current obstacles. So each team gets a fair distance.

The admin can adjust the probability of near or far delivery points for each player, so the game can be made harder or easier for a certain player.

2.6 Obstacles

On the board, red always means “this road slows you”. The light pattern shows the type of the obstacle. Team colors are far from red, so they are easy to separate. The admin can change the effect of each obstacle type.

Obstacle	Effect in the game	Look on the LED board
Heavy traffic	More NPC cars on this road, low speed	Dim red light, slow wave
Road work	Road is closed with a barrier	Bright, steady red light
Accident	One lane is closed, slow zone around it	Red and white blinking light
Water puddle	No slowing; the motorcycle slides	Cool white shimmering light

2.7 Earthquake mode

An earthquake can be triggered by hitting the board with a hand, or by the admin with a button. A gyroscope and accelerometer sensor below the board detects how strong the hit was and finds the approximate area of the hit. The admin can also start an earthquake with vibration motors under the board, and the sensor detects this shaking in the same way. The strength of the earthquake is an admin parameter. The admin can change it at any time during the game and watch the resulting seismic graphs on the admin panel.

In the game, the camera starts shaking, motorcycles fall, and roads around the center of the earthquake are blocked with roadwork. On the LED board, a shock wave moves out from this area. A bigger hit gives a bigger earthquake.

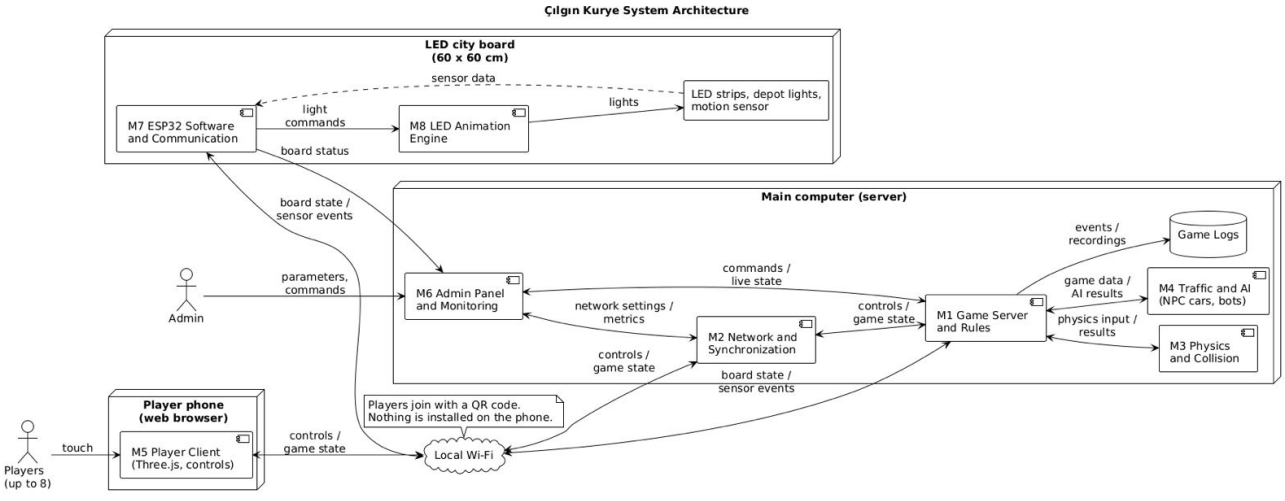
3. Physical Board

The board is 60×60 cm. The road grid is made of 6 roads in each direction. So there are 5×5 city blocks. All roads intersect at 90°, and an outer road around the city connects all four corners. The city center is in the middle of the board and has tall buildings. Further out from the center, the city is less busy, with houses and trees. The buildings are made with a 3D printer.

All roads have addressable LED strips. A motorcycle is shown as a bright dot with a small tail, so its direction of movement can be seen. In each corner there is a store with 2 RGB LEDs inside. If a team has 2 players, each LED shows the tone of one player. If a team has only one player both LEDs show the same tone. An ESP32 controls all LEDs, vibration motors and also reads the motion sensor. It is connected to the main computer over Wi-Fi.

The 3D city is made from the same city plan as the physical board, so they always match. This is important because players use only the board to navigate.

4. System Architecture



Everything related to the game runs on the main computer, and this computer makes all final decisions. The phones only send controls and display the 3D city, so everyone sees the same game. On the phone, players see the movement of their own motorcycle instantly. The LED board gets the game state from the main computer and creates the light animations itself.

5. Admin Panel: Parameters and Results

The admin can adjust all parameters below during the game. The last column shows the live graphs that the admin sees on the panel.

Group	Parameters and commands	Graphs on the admin panel
Game	Game time, number of bots	Score of each team over time
Obstacles	Add or remove an obstacle on any road, put an obstacle in front of a player, close a road, slowing amount of each obstacle type, number of NPC cars on a road	Mean delivery time, number of path changes
Players	Probability of near or far delivery points, player speed, fall penalty time	Delivery distance and score of each player
Physics	Max speed, acceleration, brake force, grip, speed limit for falls	Speed of each player, number of falls
Earthquake	Start (by button or vibration motors), magnitude	Seismic graph from the sensor, number of falls
Network (per player)	Added delay (ping), packet loss, frame rate limit on the phone	Ping, packet loss and frame rate of each player
Board Connection	Delay and packet loss on the LED board connection	Delay of the LED board behind the game

6. Modules

There are 8 modules in the project. Each module has a clear set of inputs and outputs. Each module can be developed and tested separately, with a simple mock up data of the others.

M1 Game Server and Rules

Task	Runs the main game: game time, teams, stores, deliveries, scores and falls. Selects delivery points and performs admin commands.
Input	Player controls, physics output, traffic and bot data, admin commands, earthquake events.
Output	Game state for the phones, the admin panel and the LED board.

M2 Network and Synchronization

Task	Links the phones to the main computer, synchronizes all players, and applies the network parameters (delay, packet loss) set by the admin.
Input	Controls from the phones, game state from M1, network parameters from M6.
Output	Controls to M1, game state to the phones, ping and packet loss values to M6.

M3 Physics and Collision

Task	Motion of the motorcycle (speed, brake, steering, U turn, sliding), collisions with buildings, motorcycles and cars, falls, effects of obstacles and earthquakes.
Input	Player controls, city plan, obstacles, NPC car positions from M4, earthquake events, physics parameters.
Output	New positions of the motorcycles, collision and fall events.

M4 Traffic and AI

Task	Finds paths on the city roads, moves the NPC cars, and drives the AI bots when there are not enough players.
Input	City plan, obstacles, game state, traffic and bot parameters.
Output	NPC car positions, bot controls, travel times for delivery point selection.

M5 Player Client and 3D City

Task	Web page on the phone: low poly 3D city, motorcycles and cars, game controls, compass, “Find me” button, team selection, 3D view
Input	Game state from M2, touch input on the phone.
Output	Player controls, “Find me” requests, team selection.

M6 Admin Panel and Monitoring

Task	Admin screen on the main computer: live map, obstacle tools, all parameters in Section 5, live graphs
Input	Game state and measured values from M1 and M2, LED board state and sensor data from M7.
Output	Sending parameter changes and commands.

M7 ESP32 Software and Communication

Task	Wi-Fi link between the board and the main computer, reading the motion sensor, finding the strength and area of a hit, driving the vibration motors.
Input	Game state from the main computer, motion sensor data.
Output	Light and vibration commands to M8, earthquake events to the main computer, board state and sensor data to M6.

M8 LED Animation and Physical Board

Task	Light animations (motorcycles, delivery points, obstacles, shock wave, store lights) and building the board: base, roads, LED strips, cables, power, vibration motors, 3D printed buildings.
Input	Light and vibration commands from M7, city plan.
Output	Lights on the board.

The table below shows the members of each module. Every module has at least 3 members, and every member works in 2 or 3 modules. In this way, each module can continue even if some members leave the project.

Table: Member and module matrix (x = member)

Member	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**
Tuğberk Akbulut	x	x					x	
Abdullatif Yılmaz	x		x		x			
Hasan Alperen Elmas			x	x			x	
Batuhan Yanar			x	x		x		
Ayşe Poyraz		x				x		x
Berk Andaçgil	x				x	x		
Hasan Hüseyin Özdemir		x					x	x
Muhib Taha Boy				x			x	x
Selim Barut	x		x		x			
Umutcan Ocak	x	x						x
People in module	5	4	4	3	3	3	4	4

7. Final Overview



Figure 1. Final state of the project and physical LED city board.